Andrew Hobbs

Education

PhD Agricultural and Resource Economics, *University of California, Davis* 2015-2020
Fields: Development Economics, Environmental Economics.

M.A. Economics, *University of British Columbia* 2010

Honors B.S. Economics, B.A. International Studies, *University of Utah* 2009

Employment

Assistant Professor, *Economics, University of San Francisco* 2020-Present

Graduate Student Researcher, *University of California, Davis* 2016-2020

Graduate Student Instructor, *University of California, Davis* 2016

Analyst, *Climate Policy Initiative* 2010-2015

Peer-Reviewed Publications

"Insuring the Family or the Asset? Framing Effects on Insurance Demand" (2026). *The Geneva Papers on Risk and Insurance – Issues and Practice*. Published online November 2026. Available at: `https://doi.org/10.1057/s41288-026-00403-5`

"Gender and Culture Shape Prosociality More than Heat Stress in a Five-Country Experiment" (2026). *PNAS Nexus*, 5(3), pgag017. with Alessandra Cassar, Jesse Anttila-Hughes, Francesco Bogliacino, Travis Lybbert, Irvin Rojas, Kelvin Mashisia Shikuku, John Chetwynd, Jake Cosgrove, Alexander Courtman, Andrew Hall, Stephanie Hermoso, Scott Klaus, Antonia Sottile, Nikita Tkachenko, and Bruce Wydick.

"Refractive Error Correction and Harvest Worker Productivity in the Guatemalan Coffee Sector: A Quasi-experimental Analysis" (2026). *Global Public Health*, 21(1). with Bryce Everett, Brad Wong, Erwin Herdocia, Heidy Linares, Ana Marroquín, Juan Francisco Yee, K. Davina Frick, Travis Lybbert, and Bruce Wydick.

"Insuring those who bear the risk: The impact of gender-inclusive framing on insurance uptake in Kenya" (2026). *World Development*, 201, 107330. with Julian Arteaga and Michael R. Carter.

"Get in the Zone: The Risk-Adjusted Welfare Effects of Data-Driven vs. Administrative Borders for Index Insurance Zones" (2026). *Journal of Development Economics*, 180, 103658 (online October 2025). with Ella Kirchner, Elinor Benami, Michael R. Carter, and Zhenong Jin.

"Bridging Science and Practice to En(in)sure Resilience in a Changing Climate" (2024). *Journal of Catastrophe Risk and Resilience*. with Colby K. Fisher, Elinor Benami, Michaela Dolk, Jane W. Baldwin, Inbal Becker-Reshef, Rosa I. Cuppari, Tobias Dalhaus, Gregor Leckebusch, Peter Lacovara, Adam H. Sobel, and Elizabeth Tellman. Available at: `https://journalofcrr.com/comment/02-01-fisher-et-al/`

"The Distribution of Energy Efficiency and Regional Inequality" (2023). *The Energy Journal*. Awarded the Campbell Watkins Best Paper Award, *The Energy Journal* (2023). with Puja Singhal.

"Uniting remote sensing, crop modelling and economics for agricultural risk management" (2021). *Nature Reviews Earth & Environment*. with Elinor Benami, Zhenong Jin, Michael R. Carter, Aniruddha Ghosh, Robert J. Hijmans, Benson Kenduiywo, and David B. Lobell.

"Gender Differences in Authorship of Clinical Problem-solving Articles" (2020). *Journal of Hospital Medicine* 15 (8). with Elizabeth Adler, Jennifer Babik, and Gurpreet Dhaliwal.

"Measuring the effect of bi-directional migration remittances on poverty and inequality in Nicaragua." (2012). *Applied Economics*, Volume 44, Issue 19.

Working Papers

"Financial Literacy and Entrepreneurial Aspirations: Results from Pilot Experiments in India, Peru, and Uganda" (2025). with Ester Agasha, Akash Shaji, and Bruce Wydick.

"With big data come big problems: pitfalls in measuring basis risk for crop index insurance" (2022). with Matthieu Stigler, Apratim Dey, and David B. Lobell.

Research In Progress

"How Nighttime Light Impacts Crime: The Case of Blackouts in South Africa" with Arman Khachiyan and Bhavesh Ram.

"Dynamic Household Risk Sharing with Non-Cooperative Public Goods Provision"

"Learning About Insurance" with Nathaniel Jensen and Yibekal Tessema.

"Do Droughts Drive Deforestation? Evidence from Mozambique" with Maanas Nukala.

"Impact of Agricultural Policies on Crop Residue Burning in India" with Kajal Gulati, Annoushqa Bobde, and Angira Shukla.

"Scalable Country-Level Crop Yield Modeling for Food Security and Risk Mitigation" with Jesse Anttila-Hughes, Addi Joof, Benson Adomako, James Burrell, and Alexander Flores.

Reports and Other Publications

"Gender Analysis of Pastoral Systems in Three Sub-Saharan African Countries: Evidence and Programmatic Recommendations" (2024). *Gates Gender Equality Toolbox*. with Renee Giovarelli and Naisula Lepariyo.

"Colorism and Economic Opportunity in Nigeria" (2021). *Pierless Bridges*. with Mutiu Fakorede and Airah Balogun.

"Survey of data on gender and livestock" (2020). *International Livestock Research Institute*. with Fiona Onyango.

"Satellite-based Prediction of Forage Conditions for Livestock in Northern Kenya" (2020). *ICLR 2020 Workshop on Computer Vision for Agriculture (CV4A)*. with Stacey Svetlichnaya.

"Challenges and Opportunities for Efficient Land Use in Mozambique: Taxes, Financing, and Infrastructure" (2016). *Climate Policy Initiative*. with Amanda de Albuquerque.

"San Giorgio Group Case Study: Eskom CSP 100MW plant, South Africa" (2014). *Climate Policy Initiative*. with Rodney Boyd and Anja Rosenberg.

"Improving Solar Policy: Lessons from the solar leasing boom in California." (2013). *Climate Policy Initiative*. with Elinor Benami, Brendan Pierpont, and Uday Varadarajan.

"Building Energy Efficiency in China, Germany, and the United States." (2012). *Climate Policy Initiative*. with Hermann Amecke, Aleksandra Novikova, Yang Xiu, and Shengyuan Zhang.

"Clearing the Air: Carbon Pricing and Local Air Pollution in California." (2012). *Climate Policy Initiative*. with Kath Rowley.

"Codes to Cleaner Buildings: Effectiveness of U.S. Building Energy Codes." (2011). *Climate Policy Initiative*. with Jeff Deason.

Presentations

"Evading the Eye in the Sky: Crop Residue Burning and Satellite-Based Environmental Enforcement in India" Santa Clara University Department of Economics Seminar. April 15, 2026

"Scalable Country-Level Crop Yield Modeling for Food Security and Risk Mitigation" Poster Presentation at Tackling Climate Change with Machine Learning Workshop, NeurIPS 2025. December 2025

"Mapping Global Climate Risk in Agriculture" Presentation at Integral Ecology in Action: A Decade of Laudato si', University of San Francisco. April 10, 2025

"Enabling Technology Adoption through Learning" USAID Feed the Future Innovation Lab for Markets, Risk and Resilience Webinar. November 7, 2024

"Reformulating Index Insurance to Protect Women's Assets and Well-being: Evidence from Pastoralist Communities in Kenya." IPA/GPRL Annual Researcher Gathering (September 5, 2024); UC Davis Development Economics Seminar (November 28, 2023); International Food Policy Research Institute Invited Seminar (July 20, 2023).

"Machine Learning with Satellite and Survey Data for Index Insurance." Risk and Insurance Studies Centre Climate Risk Day (March 23, 2023); American Geophysical Union Annual Meeting (December 15, 2022).

"Gender and the Decision to Insure." International Conference on Inclusive Insurance. October 27, 2022

"Machine Learning for Index Insurance." Agricultural and Applied Economics Association Annual Meeting (August 2, 2022); American Geophysical Union Annual Meeting (December 13, 2021).

"Intrahousehold Risk Sharing: Evidence from Samburu County, Kenya." American Risk and Insurance Association Annual Meeting. August 3, 2021

"Do Droughts Drive Deforestation? Evidence from Mozambique." Agricultural and Applied Economics Association Annual Meeting (August 2, 2021); Association of Environmental and Resource Economists Summer Conference (May 30, 2019).

"Artificial Intelligence for Food Security." IndabaX Tanzania. June 19, 2021

"On Heat Demand and the Distribution of Energy Efficiency." Midwest Economics Association. March 22, 2021

"Machine Learning for Optimal Index Insurance." American Geophysical Union Annual Meeting. December 15, 2020

"Household Decisions Under Risk: Evidence from Samburu County, Kenya." Agricultural and Applied Economics Association Annual Meeting (July 21, 2019, Atlanta); University of San Francisco Department of Economics Departmental Seminar (January 31, 2019).

"Durable Empowerment: Designing Insurance to Maximize Benefits for Women" Digital Credit Observatory Women's Economic Empowerment Convening, June 27, 2019, Berkeley, California.

"Computer Vision for Drought Resilience" Computer Vision and Pattern Recognition (CVPR 2019), June 15, 2019, Long Beach, California.

"More frequent flooding accelerates deforestation in Mozambique." Geo4Dev, November 15, 2018, Berkeley, California.

"Learning about pastoralist decisionmaking with games." International Center for Evaluation & Development Evidence to Action Conference, July 24, 2018, Nairobi, Kenya.

"Lessons for South Africa from the solar leasing boom in California." Solar Indaba, September 4, 2013, Cape Town, South Africa.

"Financial Incentives for Energy Efficient Retrofits in Buildings." ACEEE Buildings Summer Study, August 12, 2012.

Grants and Awards

Campbell Watkins Best Paper Award, *The Energy Journal* (2023).

Colorism and Economic Opportunity in Nigeria (2021), $5000, University of San Francisco Jesuit Foundation Grant.

Measuring Global Agricultural Weather Risk (2020), $3824, University of San Francisco Faculty Development Fund.

Sokoni: A Digital Livestock Marketplace for Northern Kenya (2019), $2900, Poverty Alleviation through Sustainable Solutions (PASS) Grant, Blum Center for Developing Economies.

Sokoni: A Digital Livestock Marketplace for Northern Kenya (2019), $3000, Henry A. Jastro Research Grant.

Droughtwatch: Computer Vision for Index Insurance (2019), Computer Vision for Global Challenges Travel Grant to CVPR 2019, Facebook.

Software and Public Tools

sdidevent: R package for the event-study decomposition of synthetic difference-in-differences (Ciccia, Clarke, and Pailañir 2024).

seliv: R package implementing the split-sample select-and-interact 2SLS estimator of Abadie, Gu, and Shen (2024, *Journal of Econometrics*).

Africa Yields: interactive web map of subnational maize yield predictions across 38 sub-Saharan African countries.

SimPastoralist: browser-based pastoralist simulation game used as the experimental instrument in *Insuring the Family or the Asset?*

Financial Literacy Game: multilingual mobile financial-literacy game with AI tutoring, used in a randomized field experiment in Uganda, India, and Peru.

Mirare: Android livestock marketplace for pastoralists in Samburu County, Kenya.

California Carbon Dashboard: live tracker of California cap-and-trade prices and emissions policy.

Teaching

Courses Taught, *University of San Francisco* 2019-Present

Graduate

ECON 640 Institutions, Markets, and Platforms
ECON 641 Microeconomics for the Digitized Economy
ECON 660 Environmental Economics
ECON 672 Economics of Development
ECON 673 Development Microeconomics
ECON 690 Development Economics Seminar
ECON 691 Economics of AI
ECON 692 Applied Economics Seminar

Undergraduate

ECON 111 Principles of Microeconomics
ECON 390 Special Topics

Advising and Curriculum, *University of San Francisco* 2020-Present

First advisor for 26 completed Master's theses (2021–2026) in the MSAE and MSIDEC programs; supervised 15 internship and work-study sections.

Designed ECON 641 and ECON 660; hired industry adjuncts from Netflix and Google; added undergraduate sections to grow the 4+1 BS–MS pipeline.

Service and Professional Activities

Academic Director, Master of Science in Applied Economics, University of San Francisco.

Co-organizer, Bay Area Tech Economics Seminar, University of San Francisco.

Affiliate, Center for Effective Global Action (CEGA), University of California, Berkeley.

Member, Partnership on AI working group on AI for Global Food Security.

Referee: American Journal of Agricultural Economics, Agricultural Economics, Economic Development and Cultural Change, European Economic Review, Journal of the Association of Environmental and Resource Economists, Journal of Development Studies, Natural Hazards, Sustainability, Geneva Papers on Risk and Insurance, Risk Management and Insurance Review, Climate Policy, Climate Research, Food Policy.

Grant proposal reviewer: USAID Feed the Future Innovation Lab (2019–2021); Foundation for Food and Agriculture Research (2022–2023); Gates Foundation (2025).